

EXERCISE 13

Detection of real-time threats, analyse recorded traffic files and identify anomalies

Aim: To learn how to use Snort for real-time threat detection, traffic analysis, and anomaly identification using both live and recorded network data.



Snort

Learn how to use Snort to detect real-time threats, analyse recorded traffic files and identify anomalies.

Medium 120 min

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Room completed (100%)

- Task 1 Introduction
- Task 2 Interactive Material and VM
- Task 3 Introduction to IDS/IPS
- Task 4 First Interaction with Snort
- Task 5 Operation Mode 1: Sniffer Mode
- Task 6 Operation Mode 2: Packet Logger Mode
- Task 7 Operation Mode 3: IDS/IPS
- Task 8 Operation Mode 4: PCAP Investigation
- Task 9 Snort Rule Structure
- Task 10 Snort2 Operation Logic: Points to Remember
- Task 11 Conclusion

Navigate to the Task-Exercises folder and run the command `"/.easy.sh"` and write the output

Too Easy! ✓ Correct Answer

Run the Snort instance and check the build number.

149 ✓ Correct Answer ? Hint

Test the current instance with `"/etc/snort/snort.conf"` file and check how many rules are loaded with the current build.

4151 ✓ Correct Answer ? Hint

Test the current instance with `"/etc/snort/snortv2.conf"` file and check how many rules are loaded with the current build.

1 ✓ Correct Answer ? Hint

What is the number of the detected HTTP GET methods?

2

✓ Correct Answer

🔍 Hint

Which IDS or IPS type can help you stop the threats on a local machine?

HIPS

✓ Correct Answer

Which IDS or IPS type can help you detect threats on a local network?

NIDS

✓ Correct Answer

Which IDS or IPS type can help you detect the threats on a local machine?

HIDS

✓ Correct Answer

Which IDS or IPS type can help you stop the threats on a local network?

NIPS

✓ Correct Answer

Which described solution works by detecting anomalies in the network?

NBA

✓ Correct Answer

According to the official description of the snort, what kind of NIPS is it?

full-blown

✓ Correct Answer

NBA training period is also known as ...

baselining

✓ Correct Answer

Investigate the traffic with the default configuration file **with ASCII mode**.

```
sudo snort -dev -K ASCII -l .
```

Execute the traffic generator script and choose **"TASK-6 Exercise"**. Wait until the traffic ends, then stop the Snort instance. Now analyse the output summary and answer the question.

```
sudo ./traffic-generator.sh
```

Now, you should have the logs in the current directory. Navigate to folder **"145.254.160.237"**. What is the source port used to connect port 53?

3009

✓ Correct Answer

🔍 Hint

Use **snort.log.1640048004**

Read the snort.log file with Snort; what is the IP ID of the 10th packet?

```
snort -r snort.log.1640048004 -n 10
```

49313

✓ Correct Answer

🔍 Hint

Read the **"snort.log.1640048004"** file with Snort; what is the referer of the 4th packet?

http://www.ethereal.com/development.html

✓ Correct Answer

🔍 Hint

Read the **"snort.log.1640048004"** file with Snort; what is the Ack number of the 8th packet?

0x38AFFFF3

✓ Correct Answer

Read the **"snort.log.1640048004"** file with Snort; what is the number of the **"TCP port 80"** packets?

41

✓ Correct Answer

🔍 Hint

Investigate the **mx-1.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

170 ✓ Correct Answer

Keep reading the output. How many TCP Segments are Queued?

18 ✓ Correct Answer

Keep reading the output. How many "HTTP response headers" were extracted?

3 ✓ Correct Answer

Investigate the **mx-1.pcap** file with the **second** configuration file.

```
sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

68 ✓ Correct Answer

Investigate the **mx-2.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-2.pcap
```

What is the number of the generated alerts?

340 ✓ Correct Answer ? Hint

Keep reading the output. What is the number of the detected TCP packets?

82 ✓ Correct Answer

Investigate the **mx-2.pcap** and **mx-3.pcap** files with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . --pcap-list="mx-2.pcap mx-3.pcap"
```

What is the number of the generated alerts?

1020 ✓ Correct Answer

Use "task9.pcap". Write a rule to filter IP ID "35369" and run it against the given pcap file. What is the request name of the detected packet? You may use this command: "snort -c local.rules -A full -l . -r task9.pcap"

TIMESTAMP REQUEST ✓ Correct Answer ? Hint

Clear the previous alert file and comment out the old rules. Create a rule to filter packets with **Syn** flag and run it against the given pcap file. What is the number of detected packets?

1 ✓ Correct Answer

Clear the previous alert file and comment out the old rules. Write a rule to filter packets with **Push-Ack** flags and run it against the given pcap file. What is the number of detected packets?

216 ✓ Correct Answer

Clear the previous alert file and comment out the old rules. Create a rule to filter **UDP** packets with the same source and destination IP and run it against the given pcap file. What is the number of packets that show the same source and destination address?

7 ✓ Correct Answer

Case Example - An analyst modified an existing rule successfully. Which rule option must the analyst change after the implementation?

rev ✓ Correct Answer

Result: Successfully configured Snort to detect live threats, analyzed pcap traffic files, and identified suspicious patterns and anomalies in network behavior.